

3.6.1 Built-in Libraries

math حساب کے لیے، random بے ترتیب عدد کے لیے اور datetime تاریخ وقت کے لیے بنی ہوئی لائبریریاں ہیں۔

CS XI

Chapter 3: Programming Fundamentals

Lecture 22 of 24

3.6.1 Built-in Libraries

LEARN IT

Built-in libraries come pre-installed with Python. You only import them; you do not run `pip`. Three that every Class XI student must know are `math`, `random` and `datetime`.

math

```
import math
print(math.pi)
print(math.ceil(4.3))
print(math.floor(4.9))
print(math.pow(2, 4))
print(math.sqrt(81))
```

OUTPUT

```
3.141592653589793
5
4
16.0
9.0
```

random and datetime

```
import random
print(random.randint(1, 1))
print(0.0 ≤ random.random() < 1.0)

import datetime
now = datetime.datetime(2026, 9, 27, 11, 59, 40)
print(now.strftime("%d-%m-%Y"))
```

OUTPUT

```
1
True
27-09-2026
```

`randint(a, b)` includes both ends. `random()` is a float in `[0.0, 1.0)`. `strftime` formats a date; `%d-%m-%Y` is day-month-year.

REMEMBER IT

`ceil` always rounds **up** (ceiling); `floor` always rounds **down**. `round()` (built-in, not in `math`) goes to the nearest integer and is a different function.

اردو میں سمجھیں

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CHECK YOURSELF

1. What does `math.ceil(4.0)` return?
2. Write a line that prints a random integer from 1 to 6 (a dice).

Answers: 1. 4 (already an integer). 2. `print(random.randint(1, 6))` after `import random`.

Next lecture: 3.6.2 Third-Party Libraries